

DESIGN AND DEVELOPMENT OF A PROGRAMMABLE LINEAR DC ELECTRONIC LOAD

ESP32-WROOM-32 Based Two-Layer PCB Prototype

Engineering Design and Mathematical Documentation

Final-Year Electrical Engineering Project

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1 Abstract

This document presents the first-pass engineering design of a programmable linear DC electronic load intended as a compact laboratory test instrument for DC power sources and batteries. The product receives electrical power from an external DC source and electronically controls the amount of current absorbed by a MOSFET power stage. The project does not use a DC–DC converter, an AC–DC converter, an inverter, solar-energy conversion, or an Internet-of-Things subsystem.

The proposed prototype is designed for a nominal input range of 10–15 V, a controlled current range of approximately 0.1–2 A, and a maximum electrical loading target of 30 W. Four BUZ11 N-channel power MOSFETs are used in parallel and operated in the linear region so that the input electrical power is intentionally dissipated as heat. Current is sensed using a $0.01\ \Omega$ shunt and an INA180A3 current-sense amplifier with a nominal gain of 100 V/V. A 12-bit MCP4725 DAC generates the commanded current reference. An LM358B analog feedback stage performs the fast current regulation, while an ESP32-WROOM-32 module, soldered directly onto the custom two-layer PCB, provides the user interface, measurement, supervisory logic, protection logic, and display control.

The user operates the instrument using a rotary encoder, push buttons, and an OLED display. The proposed operating modes are constant current, constant power, and constant resistance. A battery discharge and capacity-test application can also use the same hardware. The design intentionally uses moderate current and power so that the high-current paths are practical on a two-layer PCB. The most important physical validation tasks are MOSFET linear safe-operating-area verification, thermal characterization, control-loop stability testing, current-sensing calibration, and verification of the exact components purchased from the selected supplier.

2 Introduction

A DC source such as a laboratory power supply or battery must often be tested under a known electrical load. A conventional resistor can provide a load, but its current is determined by Ohm’s law and changes directly with source voltage. A rheostat allows manual adjustment of the resistance, but it still requires the operator to make changes when the source operating point changes.

A programmable electronic load is an active power sink. It uses a power semiconductor as a controllable element and feedback to produce a desired current or power characteristic.

The proposed project is therefore an instrumentation-oriented power-electronics product. The power stage consists of MOSFETs operating in their linear region. The controller and measurement subsystem is implemented with an ESP32-WROOM-32 mounted directly on the PCB. The final PCB is intended to look like a self-contained laboratory instrument rather than a development-board assembly.

3 Project Objectives

The objectives of the project are:

1. Design a programmable linear DC electronic load.
2. Implement the design on a custom two-layer PCB.
3. Solder the ESP32-WROOM-32 module directly to the PCB.
4. Use power MOSFETs as the active load element.
5. Implement closed-loop constant-current regulation.
6. Implement constant-power operation through supervisory digital control.

7. Implement constant-resistance emulation.
8. Measure source voltage, load current, power, and temperature.
9. Provide an OLED-based user interface.
10. Provide rotary-encoder input and push-button control.
11. Provide thermal protection and current/power limiting.
12. Provide a battery-discharge measurement application.
13. Maintain an all-inclusive project budget target below approximately INR 10,000.

4 Scope and Exclusions

The first prototype explicitly excludes the following:

1. Solar energy conversion.
2. Internet-of-Things or cloud communication.
3. DC–DC power conversion as the main power-processing function.
4. AC–DC conversion.
5. Inverter operation.
6. Circuit-breaker or electronic-fuse functionality.

The control electronics may use linear voltage regulators for their local supply rails. This does not constitute the power-processing function of the project.

5 Difference Between a Normal Resistor, Rheostat, and Electronic Load

A conventional resistor implements

$$I = \frac{V}{R}. \quad (1)$$

For a fixed resistor, the resistance does not change automatically. For a rheostat, resistance may be adjusted manually, but the terminal behavior is still approximately

$$I = \frac{V}{R}. \quad (2)$$

The electronic load instead uses an active control loop to force a desired operating characteristic.

In constant-current operation,

$$I_{\text{LOAD}} \approx I_{\text{SET}}. \quad (3)$$

In constant-power operation,

$$P_{\text{LOAD}} \approx P_{\text{SET}}, \quad (4)$$

with the required current approximately

$$I_{\text{SET}} = \frac{P_{\text{SET}}}{V_{\text{IN}}}. \quad (5)$$

In constant-resistance operation,

$$R_{\text{EQ}} = \frac{V_{\text{IN}}}{I_{\text{LOAD}}}. \quad (6)$$

The fundamental distinction is therefore that a resistor provides a physical resistance whereas the proposed product provides a programmable and regulated electrical characteristic.

5.1 Numerical Comparison

For a $10\ \Omega$ resistor connected to a $24\ \text{V}$ source,

$$I = \frac{24}{10} = 2.4\ \text{A} \quad (7)$$

and

$$P = VI = 24(2.4) = 57.6\ \text{W}. \quad (8)$$

If the source falls to $12\ \text{V}$,

$$I = \frac{12}{10} = 1.2\ \text{A}. \quad (9)$$

The resistor cannot preserve a $2.4\ \text{A}$ load.

With a programmable electronic load in constant-current mode, a $2\ \text{A}$ command can remain approximately $2\ \text{A}$ over the valid operating range.

6 Product Description

The proposed product is a small laboratory instrument that can be connected between an external DC source and the instrument input terminals.

The user provides:

- a DC source voltage;
- the desired operating mode;
- a current, power, or resistance setpoint;
- a start or stop command.

The product provides:

- controlled electrical loading;
- measured voltage;

- measured current;
- calculated power;
- measured temperature;
- protection action;
- visual information through the OLED.

7 End-to-End Workflow

7.1 Step 1: Apply External DC Power

The input is supplied by an external regulated or laboratory DC supply.

The target design range is

$$V_{IN} = 10 - -15 \text{ V.} \quad (10)$$

The preferred initial development source is a current-limited laboratory power supply or an equivalent regulated adapter capable of at least 2 A.

7.2 Step 2: Connect the Source to the PCB

The positive source terminal is connected to the PCB input positive terminal and the negative source terminal is connected to the PCB input negative terminal.

The electrical path is intentionally routed through the main current shunt and MOSFET power stage.

7.3 Step 3: Turn on the Product

The ESP32 starts in a safe state. The MOSFET command defaults to zero so the load remains effectively disabled until the system completes its self-check.

7.4 Step 4: User Selects a Mode

The user selects one of the following:

- a. CC: constant current;
- b. CP: constant power;
- c. CR: constant resistance.

7.5 Step 5: User Enters the Setpoint

The rotary encoder is rotated to change the desired value. Pressing the encoder confirms the selection.

Examples are:

$$I_{SET} = 1.5 \text{ A,} \quad (11)$$

$$P_{\text{SET}} = 20 \text{ W}, \quad (12)$$

or

$$R_{\text{SET}} = 15 \, \Omega. \quad (13)$$

7.6 Step 6: Reference Generation

The ESP32 writes the setpoint to the MCP4725 DAC. The DAC creates the analog control reference.

7.7 Step 7: Closed-Loop Regulation

The current-sense circuit measures the actual load current. The LM358B compares the desired reference against the measured current and adjusts the MOSFET gate command.

7.8 Step 8: Power Absorption

The MOSFETs operate in the linear region and dissipate the source power as heat.

At the design maximum,

$$P_{\text{IN}} = V_{\text{IN}} I_{\text{LOAD}} = 15 \text{ V} \times 2 \text{ A} = 30 \text{ W}. \quad (14)$$

7.9 Step 9: Measurement and Display

The ESP32 reads voltage, current, and temperature and calculates power using

$$P_{\text{LOAD}} = V_{\text{IN}} I_{\text{LOAD}}. \quad (15)$$

The OLED displays the measured state.

8 System and Subsystem “Jigsaw Puzzle”

The complete product is divided into the following major systems.

8.1 System A: Input and Protection

Purpose: Connect the external DC source and provide basic protection.

Parts:

- two-pin input terminal;
- input fuse;
- input bulk capacitor;
- input ceramic decoupling capacitor;
- optional reverse-polarity protection;
- optional transient protection.

8.2 System B: MOSFET Power Stage

Purpose: Absorb and control the electrical load power.

Parts:

- four BUZ11 MOSFETs;
- four source ballast resistors;
- four gate resistors;
- four gate-source pull-down resistors;
- four gate-source protection zeners;
- common heatsink;
- cooling fan.

8.3 System C: Current Measurement

Purpose: Convert load current into a measurement and feedback signal.

Parts:

- $0.01\ \Omega$ shunt;
- INA180A3 current-sense amplifier;
- current-sense filter;
- Kelvin-sense PCB routing.

8.4 System D: Analog Control Loop

Purpose: Provide fast current feedback.

Parts:

- LM358B;
- MCP4725 DAC reference;
- RC compensation/filtering;
- gate-drive resistors.

8.5 System E: Microcontroller

Purpose: User interface, reference generation, measurement, protection, and test management.

Parts:

- ESP32-WROOM-32;
- 3.3 V LDO;

- reset/EN network;
- GPIO0 boot network;
- UART programming header;
- local bypass capacitors.

8.6 System F: User Interface

Purpose: Human-machine interaction.

Parts:

- 0.96-inch SSD1306 OLED;
- EC11 rotary encoder;
- encoder push switch;
- START/STOP button;
- RESET button;
- status LEDs.

8.7 System G: Temperature and Cooling

Purpose: Monitor thermal condition and prevent excessive heating.

Parts:

- 10 k Ω NTC;
- 10 k Ω reference resistor;
- cooling fan;
- fan transistor;
- heatsink.

8.8 System H: Low-Voltage Rails

Purpose: Supply the electronic control section.

The first-pass architecture is

$$V_{\text{IN}} \rightarrow \text{L7805} \rightarrow 5\text{ V} \quad (16)$$

followed by

$$5\text{ V} \rightarrow 3.3\text{ V LDO} \rightarrow 3.3\text{ V}. \quad (17)$$

These regulators only power the control electronics.

9 Frozen Electrical Specification

Parameter	Prototype target
Input voltage	10–15 VDC
Maximum controlled current	2.0 A
Peak load power	30 W
Recommended initial continuous target	24 W
CC current range	0.1–2.0 A
CP range	approximately 5–30 W
CR useful range	approximately 5–150 Ω
Number of MOSFETs	4
Power MOSFET	BUZ11
Main current shunt	0.01 Ω , 3 W, 1%
Current-sense gain	100 V/V
DAC	MCP4725, 12 bit
Control op-amp	LM358B
Microcontroller	ESP32-WROOM-32
Display	0.96-inch SSD1306 OLED
PCB	two-layer FR-4
Preferred copper	2 oz if budget permits
Target PCB size	approximately 100 mm $\times$ 100 mm

Table 1: First-pass prototype specification.

10 Maximum Power Calculation

The maximum electrical loading point is

$$V_{\text{IN,max}} = 15 \text{ V} \quad (18)$$

and

$$I_{\text{MAX}} = 2 \text{ A}. \quad (19)$$

Therefore,

$$P_{\text{MAX}} = V_{\text{IN,max}} I_{\text{MAX}} = 15 \times 2 = \boxed{30 \text{ W}}. \quad (20)$$

The 30 W point is a peak design target. A continuous 30 W rating should only be claimed after thermal validation.

11 Current Shunt Design

Select

$$R_{\text{SH}} = 0.01 \Omega. \quad (21)$$

At 2 A,

$$V_{SH} = IR_{SH} = 2(0.01) = \boxed{20 \text{ mV}}. \quad (22)$$

The shunt power is

$$P_{SH} = I^2 R = 2^2(0.01) = \boxed{0.04 \text{ W}}. \quad (23)$$

If a 3 W shunt is used, the nominal power-rating margin is

$$M_P = \frac{3}{0.04} = 75. \quad (24)$$

The high margin reduces thermal stress and leaves room for transient conditions.

12 Current-Sense Amplifier

The INA180A3 is selected with a nominal gain

$$G_{CS} = 100 \text{ V/V}. \quad (25)$$

Therefore,

$$V_{SENSE} = G_{CS} V_{SH}. \quad (26)$$

Substituting the selected shunt,

$$V_{SENSE} = 100(I)(0.01) = I. \quad (27)$$

Thus the ideal transfer relationship is

$$\boxed{V_{SENSE}[\text{V}] \approx I[\text{A}]}. \quad (28)$$

The resulting nominal values are:

$$I = 0.1 \text{ A} \Rightarrow V_{SENSE} \approx 0.1 \text{ V}, \quad (29)$$

$$I = 0.5 \text{ A} \Rightarrow V_{SENSE} \approx 0.5 \text{ V}, \quad (30)$$

$$I = 1.0 \text{ A} \Rightarrow V_{SENSE} \approx 1.0 \text{ V}, \quad (31)$$

$$I = 1.5 \text{ A} \Rightarrow V_{SENSE} \approx 1.5 \text{ V}, \quad (32)$$

$$I = 2.0 \text{ A} \Rightarrow V_{SENSE} \approx 2.0 \text{ V}. \quad (33)$$

The exact error requires calibration because amplifier offset, gain error, shunt tolerance, shunt temperature coefficient, ADC error, and PCB resistance all contribute.

13 MCP4725 DAC Calculation

The MCP4725 is a 12-bit DAC.

The number of codes is

$$N = 2^{12} = 4096. \quad (34)$$

For a nominal 3.3 V supply, the ideal code step is

$$\Delta V_{\text{DAC}} = \frac{3.3}{4095} = \boxed{0.806 \text{ mV}}. \quad (35)$$

Because the selected analog loop is scaled so that

$$V_{\text{SET}} \approx I_{\text{SET}}, \quad (36)$$

the ideal command resolution is approximately

$$\boxed{\Delta I_{\text{CMD}} \approx 0.806 \text{ mA/LSB}}. \quad (37)$$

For 2 A, the DAC output is approximately 2 V. The code is therefore

$$N_{\text{MAX}} = 4095 \frac{2.0}{3.3} \approx 2482. \quad (38)$$

The firmware can clamp the DAC code to

$$\boxed{N_{\text{MAX}} = 2482}. \quad (39)$$

The digital resolution should not be interpreted as guaranteed absolute current accuracy.

14 Input Voltage Measurement

Use a resistive divider:

$$R_1 = 33 \text{ k}\Omega \quad (40)$$

and

$$R_2 = 7.5 \text{ k}\Omega. \quad (41)$$

The ADC input is

$$V_{\text{ADC}} = V_{\text{IN}} \frac{R_2}{R_1 + R_2}. \quad (42)$$

The division ratio is

$$\frac{7.5}{33 + 7.5} = 0.185185. \quad (43)$$

At 15 V,

$$V_{\text{ADC,max}} = 15(0.185185) = \boxed{2.778 \text{ V}}. \quad (44)$$

The divider current is

$$I_{\text{DIV}} = \frac{15}{33k + 7.5k} = \frac{15}{40.5k} \approx \boxed{0.370 \text{ mA}}. \quad (45)$$

The total divider power is

$$P_{\text{DIV}} = \frac{15^2}{40.5k} \approx \boxed{5.56 \text{ mW}}. \quad (46)$$

Therefore the divider does not have a significant thermal burden.

14.1 Voltage-Measurement Filter

The divider Thevenin resistance is

$$R_{\text{TH}} = R_1 \parallel R_2 \approx 6.11 \text{ k}\Omega. \quad (47)$$

Using

$$C_{\text{ADC}} = 47 \text{ nF}, \quad (48)$$

the nominal corner frequency is

$$f_c = \frac{1}{2\pi R_{\text{TH}} C_{\text{ADC}}} \approx \boxed{554 \text{ Hz}}. \quad (49)$$

15 Current-Feedback Filter

For a first-pass current-feedback filter, choose

$$R_F = 10 \text{ k}\Omega \quad (50)$$

and

$$C_F = 100 \text{ nF}. \quad (51)$$

The nominal cutoff is

$$f_{c,I} = \frac{1}{2\pi R_F C_F} = \frac{1}{2\pi(10k)(100nF)} \approx \boxed{159 \text{ Hz}}. \quad (52)$$

This filtering is intended to reject unwanted high-frequency electrical noise and help the analog loop remain stable.

The actual compensation must be verified by simulation and laboratory measurements. The calculated RC corner is not a substitute for phase-margin verification.

16 MOSFET Selection and Number

Four BUZ11 N-channel MOSFETs are proposed.

At the maximum current,

$$I_{\text{TOTAL}} = 2 \text{ A.} \quad (53)$$

For ideal equal sharing,

$$I_{\text{FET}} = \frac{2}{4} = \boxed{0.5 \text{ A}}. \quad (54)$$

The number of MOSFETs is selected mainly because of thermal and linear-mode stress considerations. If one device were used, it would need to absorb approximately 30 W.

For two devices,

$$P_{\text{FET}} \approx 15 \text{ W} \quad (55)$$

per device.

For four devices,

$$P_{\text{FET}} \approx 7.5 \text{ W} \quad (56)$$

per device.

Therefore four devices provide a more practical thermal distribution for a student prototype.

The final MOSFET choice must be validated using the manufacturer's linear safe-operating-area data at the actual V_{DS} and I_D operating point. The headline switching-current rating is not sufficient evidence for continuous linear operation.

17 Source Ballast Resistors

Use

$$R_S = 0.1 \Omega \quad (57)$$

for each MOSFET source.

At 0.5 A,

$$V_{R_S} = IR_S = 0.5(0.1) = \boxed{50 \text{ mV}}. \quad (58)$$

The resistor power is

$$P_{R_S} = I^2 R = 0.5^2(0.1) = \boxed{25 \text{ mW}}. \quad (59)$$

A 3 W resistor therefore has large power margin.

The ballast resistor introduces local negative feedback and improves current sharing between parallel MOSFETs.

18 MOSFET Power Balance

At the maximum input point,

$$P_{\text{IN}} = 30 \text{ W}. \quad (60)$$

The shunt loss is

$$P_{\text{SH}} = 0.04 \text{ W}. \quad (61)$$

The four ballast resistors dissipate approximately

$$P_{\text{BAL,total}} = 4(0.025 \text{ W}) = \boxed{0.10 \text{ W}}. \quad (62)$$

Therefore the approximate MOSFET-bank dissipation is

$$P_{\text{MOS,total}} \approx 30 - 0.04 - 0.10 = \boxed{29.86 \text{ W}}. \quad (63)$$

For four devices,

$$P_{\text{MOS,each}} \approx \frac{29.86}{4} = \boxed{7.47 \text{ W}}. \quad (64)$$

The value is an ideal equal-sharing estimate. Actual device-to-device variation must be checked experimentally.

19 Thermal Design

The dominant physical constraint of a linear electronic load is heat dissipation.

The thermal path is

$$P_{\text{MOS}} \rightarrow T_J \rightarrow T_C \rightarrow T_{\text{HS}} \rightarrow T_A. \quad (65)$$

For a preliminary calculation, assume

$$R_{\theta JC} \approx 1.67^\circ\text{C/W}. \quad (66)$$

At 7.47 W per MOSFET,

$$\Delta T_{JC} = 7.47(1.67) \approx \boxed{12.47^\circ\text{C}}. \quad (67)$$

Assuming

$$R_{\theta CS} = 0.5^\circ\text{C/W}, \quad (68)$$

the case-to-heat-sink temperature rise is

$$\Delta T_{CS} = 7.47(0.5) = \boxed{3.74^\circ\text{C}}. \quad (69)$$

For an illustrative ambient temperature of

$$T_A = 35^\circ\text{C} \quad (70)$$

and a preliminary junction target of

$$T_J = 100^\circ\text{C}, \quad (71)$$

the total junction-to-ambient thermal resistance for one branch would be

$$R_{\theta JA, \text{target}} = \frac{100 - 35}{7.47} = \boxed{8.70^\circ\text{C/W}}. \quad (72)$$

Because the MOSFETs share one heatsink, a common heatsink should be evaluated for approximately 30 W total dissipation. If a heatsink temperature of 80 °C is accepted at 35 °C ambient, the corresponding shared-heatsink target is

$$R_{\theta SA, \text{shared}} \leq \frac{80 - 35}{30} = \boxed{1.5^\circ\text{C/W}}. \quad (73)$$

This is a target for design and measurement, not a guaranteed property of a particular retail heatsink.

20 MOSFET Gate Network

For each MOSFET use:

$$R_G = 1\text{ k}\Omega, \quad (74)$$

$$R_{GS} = 100\text{ k}\Omega, \quad (75)$$

and an approximately 8.2 V gate-source zener for transient protection.

At an assumed 5 V gate-drive level, the initial resistor-limited gate current is

$$I_{G, \text{initial}} \approx \frac{5}{1k} = \boxed{5\text{ mA}} \quad (76)$$

per gate branch.

The static pull-down current is

$$I_{PD} = \frac{5}{100k} = \boxed{50\text{ }\mu\text{A}}. \quad (77)$$

The gate pull-down ensures that the MOSFET tends toward an OFF state when the control signal becomes high impedance.

21 Analog Current-Control Loop

The desired current is represented by the DAC output V_{SET} . The current-sense circuit produces V_{SENSE} .

The loop error is

$$V_E = V_{\text{SET}} - V_{\text{SENSE}}. \quad (78)$$

The analog controller adjusts the MOSFET gate voltage to reduce this error.

At steady state,

$$V_E \rightarrow 0. \quad (79)$$

Therefore,

$$V_{\text{SET}} \approx V_{\text{SENSE}}. \quad (80)$$

Using the selected sensor transfer,

$$V_{\text{SENSE}} \approx I_{\text{LOAD}}, \quad (81)$$

so

$$\boxed{I_{\text{LOAD}} \approx V_{\text{SET}}} \quad (82)$$

when the units are represented numerically as amperes and volts.

For example,

$$V_{\text{SET}} = 1.5 \text{ V} \quad (83)$$

corresponds ideally to

$$I_{\text{LOAD}} \approx 1.5 \text{ A}. \quad (84)$$

22 Constant-Current Mode

In CC mode,

$$I_{\text{CMD}} = I_{\text{SET}}. \quad (85)$$

For

$$I_{\text{SET}} = 1.5 \text{ A}, \quad (86)$$

the analog loop attempts to keep

$$I_{\text{LOAD}} \approx 1.5 \text{ A}. \quad (87)$$

At 15 V,

$$P = 15(1.5) = \boxed{22.5 \text{ W}}. \quad (88)$$

At 12 V,

$$P = 12(1.5) = \boxed{18 \text{ W}}. \quad (89)$$

The current is approximately constant while the absorbed power changes with voltage.

23 Constant-Power Mode

In CP mode, the user specifies

$$P_{\text{SET}}. \quad (90)$$

The ESP32 calculates

$$I_{\text{REQ}} = \frac{P_{\text{SET}}}{V_{\text{IN}}}. \quad (91)$$

The command must be restricted by both the maximum current and maximum power:

$$I_{\text{CMD}} = \min \left[I_{\text{MAX}}, \frac{P_{\text{MAX}}}{V_{\text{IN}}}, \frac{P_{\text{SET}}}{V_{\text{IN}}} \right]. \quad (92)$$

For

$$P_{\text{SET}} = 20 \text{ W}, \quad (93)$$

the requested currents are:

$$V_{\text{IN}} = 15 \text{ V} \Rightarrow I_{\text{REQ}} = \frac{20}{15} = 1.333 \text{ A}, \quad (94)$$

$$V_{\text{IN}} = 12 \text{ V} \Rightarrow I_{\text{REQ}} = \frac{20}{12} = 1.667 \text{ A}, \quad (95)$$

$$V_{\text{IN}} = 10 \text{ V} \Rightarrow I_{\text{REQ}} = \frac{20}{10} = 2 \text{ A}. \quad (96)$$

At voltages below 10 V, the 2 A maximum prevents true 20 W constant-power operation.

24 Constant-Resistance Mode

For a requested equivalent resistance

$$R_{\text{SET}}, \quad (97)$$

the current command is

$$I_{\text{REQ}} = \frac{V_{\text{IN}}}{R_{\text{SET}}}. \quad (98)$$

The final command is restricted by the hardware limits:

$$I_{\text{CMD}} = \min \left[I_{\text{MAX}}, \frac{P_{\text{MAX}}}{V_{\text{IN}}}, \frac{V_{\text{IN}}}{R_{\text{SET}}} \right]. \quad (99)$$

For example, at

$$V_{\text{IN}} = 15 \text{ V} \quad (100)$$

and

$$R_{\text{SET}} = 15 \, \Omega, \quad (101)$$

the ideal current is

$$I = \frac{15}{15} = 1 \, \text{A} \quad (102)$$

and the power is

$$P = 15(1) = 15 \, \text{W}. \quad (103)$$

25 Useful Equivalent Resistance Range

At the lower design voltage of 10 V and the maximum current of 2 A,

$$R_{\text{MIN}} = \frac{10}{2} = \boxed{5 \, \Omega}. \quad (104)$$

At 15 V and a practical minimum regulated current of 0.1 A,

$$R_{\text{MAX}} = \frac{15}{0.1} = \boxed{150 \, \Omega}. \quad (105)$$

Thus a reasonable first-pass specification is

$$\boxed{R_{\text{EQ}} \approx 5 - 150 \, \Omega} \quad (106)$$

subject to the simultaneous current and power constraints.

The exact useful range must be experimentally characterized.

26 Temperature Measurement

A 10 k Ω NTC with a nominal B -coefficient near 3950 K is used for temperature measurement.

At 25 $^{\circ}\text{C}$,

$$R_{\text{NTC}} \approx 10 \, \text{k}\Omega. \quad (107)$$

With a 3.3 V divider using a 10 k Ω reference resistor,

$$V_{\text{NTC}} = 3.3 \frac{10}{10 + 10} = \boxed{1.65 \, \text{V}}. \quad (108)$$

The beta-model equation is

$$R(T) = R_0 \exp \left[B \left(\frac{1}{T} - \frac{1}{T_0} \right) \right], \quad (109)$$

where T and T_0 are in kelvin.

Suggested firmware thresholds are:

$$T < 60^\circ\text{C} : \quad \text{normal cooling}, \quad (110)$$

$$60^\circ\text{C} \leq T < 75^\circ\text{C} : \quad \text{fan at high speed}, \quad (111)$$

$$75^\circ\text{C} \leq T < 85^\circ\text{C} : \quad \text{power derating}, \quad (112)$$

$$T \geq 85^\circ\text{C} : \quad \text{shutdown}. \quad (113)$$

These are engineering control thresholds and must be verified against the measured thermal behavior of the final assembly.

27 Low-Voltage Electronics Rails

The LM358B control section is powered from the input-side electronics rail. The ESP32 is powered from a regulated 3.3 V rail.

A simple first-pass control-power architecture is:

$$V_{\text{IN}} \rightarrow \text{L7805} \rightarrow 5\text{ V} \quad (114)$$

and

$$5\text{ V} \rightarrow 3.3\text{ V LDO} \rightarrow 3.3\text{ V}. \quad (115)$$

Assume a 5 V rail current of approximately 150 mA. At 15 V input, the L7805 power dissipation is

$$P_{7805} = (15 - 5)(0.15) = \boxed{1.5\text{ W}}. \quad (116)$$

Therefore the 7805 requires a suitable thermal path.

For the 3.3 V rail at approximately 150 mA,

$$P_{3.3} = (5 - 3.3)(0.15) = \boxed{0.255\text{ W}}. \quad (117)$$

This is a comparatively small thermal burden.

28 ESP32-WROOM-32 Hardware

The ESP32-WROOM-32 is soldered directly to the custom PCB.

The PCB provides:

- 3.3 V supply;
- EN/reset circuitry;
- GPIO0 boot circuitry;
- UART programming header;
- local decoupling capacitors;
- appropriate antenna keep-out area;
- continuous ground reference where appropriate.

No ESP32 development board is used.

28.1 Suggested GPIO Assignment

Function	ESP32 GPIO
OLED SDA and MCP4725 SDA	GPIO21
OLED SCL and MCP4725 SCL	GPIO22
Current ADC	GPIO35
Input-voltage ADC	GPIO34
NTC ADC	GPIO39
Encoder A	GPIO18
Encoder B	GPIO19
Encoder push	GPIO23
START/STOP	GPIO25
Fan control	GPIO26
Fault input	GPIO27
UART TX	GPIO1
UART RX	GPIO3
BOOT	GPIO0
RESET	EN

Table 2: Suggested ESP32 pin assignment.

GPIO34, GPIO35, and GPIO39 are input-only pins on the classic ESP32 and are therefore suitable for ADC measurements.

29 User Input and Display

The user interacts with the product through a rotary encoder and push controls.

The interaction sequence is:

1. Power on the instrument.
2. Select CC, CP, or CR.
3. Rotate the encoder to change the setpoint.
4. Press the encoder to confirm.
5. Press START.
6. Observe voltage, current, power, and temperature.
7. Press STOP to disable the load.

An example CC display is:

PROGRAMMABLE DC LOAD

```
VIN : 15.00 V
I   : 1.500 A
P   : 22.50 W
TEMP: 43.2 C
```

```
MODE: CC
SET  : 1.500 A
```

An example CP display is:

PROGRAMMABLE DC LOAD

VIN : 12.50 V
 I : 1.600 A
 P : 20.00 W
 TEMP: 41.9 C

MODE: CP
 SET : 20.00 W

30 Battery Discharge and Capacity-Test Application

The same hardware can perform a controlled battery-discharge test.

The user selects a discharge current and a cutoff voltage.

For example,

$$I_{\text{DISCHARGE}} = 1 \text{ A.} \quad (118)$$

The ESP32 records the voltage, current, elapsed time, charge, and energy.

The delivered charge is

$$Q = \int I(t) dt. \quad (119)$$

For sampled values,

$$Q \approx \sum_{k=1}^N I_k \Delta t. \quad (120)$$

The delivered energy is

$$E = \int V(t) I(t) dt \quad (121)$$

and numerically,

$$E \approx \sum_{k=1}^N V_k I_k \Delta t. \quad (122)$$

The displayed capacity is therefore a measured capacity under the chosen discharge current and cutoff condition.

31 Protection Strategy

The first revision should include:

1. Input fuse.

2. Default-off gate pull-downs.
3. Gate-source zener protection.
4. Current-limit clamping.
5. Power-limit clamping.
6. Thermal shutdown.
7. Fan operation.
8. Sensor plausibility checks.
9. Fault latch.

A future revision can add an independent analog over-current comparator so that a hardware path can disable the power stage without relying solely on firmware.

32 Two-Layer PCB Design

The project uses a two-layer FR-4 PCB.

The layout should be divided into:

1. user-interface region;
2. ESP32 and low-voltage electronics region;
3. current-sensing and analog-feedback region;
4. MOSFET power region;
5. input and shunt region.

The current rating is only

$$I_{\text{MAX}} = 2 \text{ A}, \tag{123}$$

which is moderate for a properly designed two-layer PCB.

The principal routing rules are:

- use wide copper for the main current path;
- keep the current path short;
- avoid narrow trace necks;
- use copper pours where appropriate;
- use via stitching if current changes layers;
- route Kelvin sense lines separately;
- keep the ESP32 and ADC area away from noisy power-node regions;
- provide sufficient copper beneath thermal components.

32.1 Kelvin Shunt Connection

At the maximum current, the useful shunt signal is only

$$V_{\text{SH}} = 20 \text{ mV}. \quad (124)$$

Therefore the current-sense lines must not share the high-current copper path unnecessarily.

A four-terminal/Kelvin concept is recommended so that the INA180 measures the intended shunt voltage rather than the voltage drop in the surrounding PCB copper and solder joints.

33 Approximate PCB Trace Resistance

For illustration, assume a 5 mm-wide, 0.035 mm-thick copper trace with a length of 100 mm.

The copper cross-sectional area is

$$A = 5(0.035) = 0.175 \text{ mm}^2. \quad (125)$$

With copper resistivity approximately

$$\rho \approx 1.724 \times 10^{-8} \Omega \text{ m}, \quad (126)$$

the approximate resistance of the trace is

$$R \approx \rho \frac{L}{A} \approx 9.85 \text{ m}\Omega. \quad (127)$$

At 2 A,

$$V_{\text{DROP}} = IR \approx 2(0.00985) = \boxed{19.7 \text{ mV}}. \quad (128)$$

The associated copper loss is

$$P_{\text{TRACE}} = I^2 R \approx 4(0.00985) = \boxed{39.4 \text{ mW}}. \quad (129)$$

This calculation illustrates that wide, short traces are desirable, especially because the trace drop can be comparable with the 20 mV full-scale shunt signal.

34 Input Supply and Development Setup

The electronic load does not generate the power being measured. The power comes from the external device under test.

During development, the preferred source is a current-limited bench power supply.

A suitable development sequence is:

1. Set the laboratory supply to 10 V.
2. Set a low current limit.
3. Verify the control electronics.

4. Increase the load gradually.
5. Increase the source voltage only after the previous step is stable.

A regulated 15 V source capable of at least 2 A can be used for the final prototype test.

The external source is therefore the input of the product, while the product's output is the controlled current sink path.

35 Design Verification

The following quantities are analytically determined:

Parameter	Calculated result	Verification
Maximum electrical power	30 W	VI measurement
Shunt voltage at 2 A	20 mV	DMM/scope
Current-sense voltage at 2 A	approximately 2 V	DMM/scope
DAC step	approximately 0.806 mV	analytical
Command resolution	approximately 0.806 mA	analytical
Ideal MOSFET branch current	0.5 A	branch-current measurement
Approximate branch dissipation	7.47 W	$V_{DS}I_D$
Voltage-divider output at 15 V	2.778 V	DMM
Current filter corner	approximately 159 Hz	RC calculation
Voltage filter corner	approximately 554 Hz	RC calculation
Thermal performance	To be measured	thermocouple/IR/NTC
MOSFET SOA margin	To be verified	manufacturer data + test
Control-loop stability	To be verified	SPICE + oscilloscope
Absolute current accuracy	To be calibrated	reference-current test

Table 3: Analytical values and required physical verification.

36 Prototype Bring-Up Procedure

The hardware must be brought up progressively.

36.1 Stage 1: Control Electronics Only

With the main MOSFETs removed, verify:

- 5 V rail;
- 3.3 V rail;
- ESP32 programming;
- OLED;
- encoder;
- DAC;
- ADC inputs;
- temperature sensor.

36.2 Stage 2: Current Measurement

Use a safe external current path to verify the shunt and current-sense amplifier.

Expected nominal values include:

$$I = 0.5 \text{ A} \Rightarrow V_{\text{SENSE}} \approx 0.5 \text{ V}, \quad (130)$$

$$I = 1.0 \text{ A} \Rightarrow V_{\text{SENSE}} \approx 1.0 \text{ V}, \quad (131)$$

$$I = 1.5 \text{ A} \Rightarrow V_{\text{SENSE}} \approx 1.5 \text{ V}. \quad (132)$$

Perform zero-current calibration first.

36.3 Stage 3: One-MOSFET Test

Install one MOSFET and operate at a low-power point:

$$V_{\text{IN}} = 10 \text{ V} \quad (133)$$

and

$$I = 0.2 \text{ A}. \quad (134)$$

Therefore,

$$P = 10(0.2) = 2 \text{ W}. \quad (135)$$

Verify loop stability and temperature.

36.4 Stage 4: Four-MOSFET Test

Install all four MOSFETs and gradually increase the operating point:

$$10 \text{ V}, 0.5 \text{ A} \quad (136)$$

then

$$12 \text{ V}, 1 \text{ A}, \quad (137)$$

then

$$15 \text{ V}, 1.5 \text{ A}, \quad (138)$$

and only then the 15 V, 2 A peak point.

36.5 Stage 5: Thermal Characterization

Measure:

- each MOSFET case temperature;

- heatsink temperature;
- PCB temperature;
- shunt temperature;
- regulator temperature;
- fan behavior.

Only after thermal validation should the final continuous-power specification be stated.

37 Firmware State Machine

A simple state machine is proposed:

1. POWER ON;
2. SELF TEST;
3. STANDBY;
4. SETTING;
5. ACTIVE;
6. FAULT;
7. RESET.

In POWER ON and SELF TEST states, the load command remains zero.

During SETTING, the user selects the operating mode and setpoint.

During ACTIVE, the ESP32 reads the sensors, computes the commanded current, updates the DAC, updates the OLED, and executes the protection rules.

During FAULT, the command is forced to zero and the user is informed of the fault condition.

38 Component Selection and Robu Availability Basis

The following parts were selected because corresponding products or product families are listed by Robu during the design study.

Component	Qty.	Role	Selection basis
ESP32-WROOM-32	1	MCU	Direct PCB-mounted module
INA180A3	1	Current sensing	100 V/V nominal gain
MCP4725A0T-E/CH	1	DAC	12-bit setpoint generation
BUZ11	4 fitted + spares	Power MOSFET bank	Linear-stage candidate
0.01 Ω shunt	1	Main current sensing	Low insertion loss
0.1 Ω ballast resistor	4	Branch balancing	Stabilizes parallel sharing

LM358B	1	Analog feedback	Low-voltage control amplifier
0.96-inch SSD1306 OLED	1	Display	Local user interface
EC11 rotary encoder	1	User input	Setpoint selection
L7805	1	5 V regulator	Controller supply
3.3 V LDO	1	ESP32 supply	3.3 V rail
10 k Ω NTC	1	Thermal sensing	Heatsink temperature
TO-220 heatsink	1	Thermal management	Common MOSFET cooling
Cooling fan	1	Forced air	Thermal management
Fuse	1–2	Input protection	Fault protection
Screw terminals	2	Power connections	Input/output connection

The exact SKU, current inventory quantity, and price must be checked again immediately before placing the order because online inventory is dynamic.

39 Robu Reference Links

1. ESP32 modules:
<https://robu.in/product-category/esp-wifi-module/>
2. MOSFET category:
<https://robu.in/product-category/mosfet/>
3. Current and voltage sensors:
<https://robu.in/product-category/current-and-voltage-sensor/>
4. Current-sense resistors:
<https://robu.in/product-category/current-sense-resistors/>
5. OLED displays:
<https://robu.in/product-category/oled-display/>
6. Aluminium heatsinks:
<https://robu.in/product-category/aluminium-heat-sink/>
7. Cooling fans:
<https://robu.in/product-category/5v-cooling-fans/>
8. PCB manufacturing service:
<https://robu.in/product/online-pcb-manufacturing-service/>

40 Project Budget

The project budget must include:

1. PCB fabrication.
2. ESP32 module.
3. Power MOSFETs and spare devices.
4. Current-sense amplifier.
5. DAC.

6. Op-amp.
7. OLED.
8. Rotary encoder and switches.
9. Shunt and ballast resistors.
10. All passive components.
11. Protection components.
12. Cooling hardware.
13. Connectors.
14. Prototype-source allowance.
15. Shipping.
16. Spare components.

A conservative planning budget is:

Category	Planning allowance (INR)
ESP32-WROOM-32	500
Six BUZ11 devices including spares	300
INA180A3 and MCP4725	200
LM358B and other control ICs	150
OLED and encoder	300
Shunt and ballast resistors	100
Other passive components	500
Connectors, fuse, buttons, LEDs	400
Cooling and heatsink	350
PCB fabrication allowance	1500
External DC source allowance	1000
Spare/contingency allowance	1500
Shipping allowance	800
Total planning allowance	INR 7,600

Table 5: Conservative budget planning target.

The project ceiling is

$$\boxed{\text{Total Project Cost} \leq \text{INR } 10,000}. \quad (139)$$

The remaining margin is intended for price variation, shipping variation, spare components, and possible PCB re-fabrication.

41 Engineering Risk Register

Risk	Possible effect	Mitigation
MOSFET thermal overload	Device failure	Four devices, heatsink, fan, power derating
Unequal MOSFET current sharing	Local hot spots	Source ballast resistors
Shunt wiring error	Current-measurement error	Kelvin routing
Sensor offset	Low-current error	Zero-current calibration
Loop oscillation	Unstable load	RC filtering and oscilloscope verification
Gate overvoltage	MOSFET damage	Gate-source zener protection
PCB hotspot	Local heating	Wide copper and short power paths
Regulator heating	Controller instability	Heat sinking and current budgeting
Device-lot variation	SOA uncertainty	Use exact manufacturer's datasheet
Insufficient heatsink	Thermal shutdown	Measure heatsink temperature

Table 6: Primary engineering risks and mitigations.

42 Final Demonstration Plan

42.1 Demonstration 1: Constant Current

Set

$$I_{\text{SET}} = 1 \text{ A.} \quad (140)$$

Apply 15 V.

The expected load power is

$$P \approx 15 \times 1 = 15 \text{ W.} \quad (141)$$

Then reduce the source to 12 V and observe whether the current remains approximately 1 A.

42.2 Demonstration 2: Constant Power

Set

$$P_{\text{SET}} = 20 \text{ W.} \quad (142)$$

At 15 V,

$$I \approx 1.333 \text{ A.} \quad (143)$$

At 12 V,

$$I \approx 1.667 \text{ A.} \quad (144)$$

The display should show approximately constant power while the current changes.

42.3 Demonstration 3: Constant Resistance

Set

$$R_{\text{SET}} = 15 \, \Omega. \quad (145)$$

At 15 V,

$$I \approx 1 \text{ A} \quad (146)$$

and at 12 V,

$$I \approx 0.8 \text{ A.} \quad (147)$$

42.4 Demonstration 4: Temperature Protection

Increase the absorbed power while observing temperature.

When

$$T \geq T_{\text{TRIP}}, \quad (148)$$

the firmware sets

$$I_{\text{CMD}} = 0. \quad (149)$$

The OLED reports the thermal fault.

43 Research Basis

Programmable electronic loads are established laboratory instruments for testing DC power supplies, batteries, adapters, and other power sources. The engineering literature includes designs using power MOSFETs as controllable current sinks, current sensing, analog feedback, digital control, and battery discharge measurement.

The literature review for the final project should focus on:

- programmable electronic load architectures;
- linear-region operation of power MOSFETs;
- current-sense amplifier design;
- closed-loop current regulation;
- battery discharge and capacity measurement;
- electro-thermal behavior of MOSFETs;

- PCB power-current and thermal design.

IEEE Xplore and peer-reviewed sources should be used to select the final references for the academic report.

44 Important Engineering Limitations

The following points must be treated as design-validation requirements rather than assumptions.

44.1 MOSFET Linear SOA

The BUZ11 headline current rating is not sufficient for this project because the MOSFET is used in its linear region. The exact manufacturer's forward-bias safe-operating-area curve must be checked at the actual voltage-current operating point.

44.2 Thermal Rating

The 30 W peak value is a calculated electrical design point. The continuous rating depends on the actual heatsink, airflow, ambient temperature, thermal interface, MOSFET package, and PCB arrangement.

44.3 Current Accuracy

The nominal relationship

$$V_{\text{SENSE}} \approx I \quad (150)$$

is an ideal transfer. Real accuracy requires calibration.

44.4 Control-Loop Stability

The selected RC values are first-pass values. The completed circuit should be simulated and then evaluated experimentally with an oscilloscope.

44.5 PCB Current Handling

A properly routed two-layer PCB is suitable for a 2 A current path, but the final trace width, copper thickness, connector selection, via structure, and thermal rise must be checked against the final PCB fabrication specification.

45 Conclusion

The proposed product is a programmable linear DC electronic load rather than a conventional fixed resistor or manually adjusted rheostat. The key engineering function is to control the current absorbed from an external DC source through a closed-loop power-semiconductor stage.

The first-pass architecture uses:

ESP32-WROOM-32 + MCP4725 + INA180A3 + LM358B + 4 × BUZ11 + Current Shunt + OLED

(151)

The principal electrical specifications are:

$$V_{\text{IN}} = 10 - -15 \text{ V} \quad (152)$$

$$I_{\text{LOAD,max}} = 2 \text{ A} \quad (153)$$

$$P_{\text{LOAD,max}} = 30 \text{ W} \quad (154)$$

and a practical constant-resistance emulation range of approximately

$$R_{\text{EQ}} \approx 5 - -150 \Omega \quad (155)$$

subject to the simultaneous voltage, current, power, and thermal limits.

The two-layer PCB is feasible because the maximum load current is only 2 A. The dominant engineering challenge is thermal dissipation and linear safe-operating-area verification of the MOSFET bank.

The design is intentionally structured so that the ESP32 is not the power converter. The ESP32 provides user interaction, setpoint generation, measurement, protection logic, and test sequencing, while the analog current loop performs the fast current control. This separation improves the power-electronics content of the project and provides a clear academic basis for discussing semiconductor operation, sensing, feedback control, thermal design, and PCB engineering.

46 References

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https://documentation.espressif.com/esp32-wroom-32_datasheet_en.html
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<https://docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/>
3. Texas Instruments, *INA180 Low- and High-Side Voltage Output Current-Sense Amplifier*.
<https://www.ti.com/product/INA180>
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<https://www.ti.com/>
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<https://ieeexplore.ieee.org/>